Maxwell's Equations from the Vortex Rope Model

A Derivation of All Four Maxwell Equations from Rope Network Vortex Dynamics

Mark Palmer · June 2026

ABSTRACT

We show that all four of Maxwell's equations follow from treating the rope network as a vortex fluid. The magnetic field is identified with the vorticity of the network flow field ($B = \nabla\times v$), and the electric field with the rope pressure gradient ($E = -\nabla\varphi$). Under this dictionary: (1) $\nabla\cdot B = 0$ follows from the vector calculus identity $\nabla\cdot(\nabla\times v) = 0$; (2) Faraday's law follows from the Helmholtz vorticity theorem; (3) Gauss's law for electricity follows from the fluid source/sink equation; (4) the Ampère-Maxwell law follows from the Navier-Stokes equation with compressibility, yielding the displacement current automatically. The wave speed $c = \sqrt{(T/\mu)}$ identifies $\mu_0\varepsilon_0 = \mu/T = 1/c^2$. This derivation reproduces Maxwell's own 1861 molecular-vortex approach and places it on the rope network geometry of the Hopf bundle S^3. The one remaining gap is the numerical values of ε_0 and μ_0 from rope parameters — equivalent to deriving $\alpha = 1/137$.

Keywords: Maxwell equations · vortex fluid · rope network · magnetic field · Helmholtz theorem · displacement current · Hopf bundle

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THE OCEAN ANALOGY

The rope network behaves like an ocean. It does three things:

1. BUILD PRESSURE. Pressure differences in the network are electric fields ($E = -\nabla\varphi$, the rope pressure gradient). A charge is a point where the network is locally compressed or rarefied.
2. FORM WHIRLPOOLS. Vortices in the network flow field are magnetic fields ($B = \geqq\times v$, the vorticity of the network flow). Not swinging ropes, not rigid rotation — whirlpools, with flow fast near the centre and slow at the edges.
3. CARRY WAVES. When pressure and whirlpools change together, the disturbance travels through the network as an electromagnetic wave at speed $c = \sqrt{(T/\mu)}$. In the GEM extension (rope_gem_equations), the same ocean carries gravitational waves at the same speed.

NOTE ON TERMINOLOGY: The papers use 'network flow field' rather than 'rope velocity field.' The flow field is a state description of the network at each point — not a statement that the physical ropes are translating through space. Individual rope molecules do not follow simple circular paths, just as individual water molecules do not trace tidy whirlpool orbits. The vorticity is a property of the field, not the trajectories.

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1. The Physical Picture

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Gaede (2014) proposed that the magnetic field consists of ropes rotating around a current-carrying wire. The mathematical analysis of this model reveals a crucial detail: the ropes do not rotate as a rigid body (where velocity would increase with distance, $v \propto r$). Instead they rotate as a vortex — like water spiralling around a drain — where velocity decreases with distance ($v \propto 1/r$). This is the only rotation profile that reproduces the observed $1/r$ dependence of the magnetic field.

This vortex picture is not an ad hoc modification. It is the natural consequence of treating the rope network as a fluid. In classical fluid dynamics, a line vortex has exactly the velocity profile $v = \Gamma/(2\pi r)$, where Γ is the circulation — the same structure as the magnetic field around a wire, $B = \mu_0 I/(2\pi r)$.

1.1 The Dictionary

Rope network quantity	**Symbol**	**Maxwell/EM quantity**	**Symbol**
Network flow field	v	Vector potential	A
Vorticity of network flow	$\nabla\times v$	Magnetic field	B
Rope pressure	φ	Electric (scalar) potential	φ
Rope pressure gradient	$-\nabla\varphi$	Electric field	E
Moving knots (current)	J_rope	Electric current density	J
Rope mass density	μ	Inverse permittivity	$1/\varepsilon_0$
Rope tension	T	Related to permeability	$1/\mu_0$
Rope wave speed	$\sqrt{(T/\mu)}$	Speed of light	c
Rope winding number	n	Electric charge	q

The identification $B = \nabla\times v$ means the magnetic field is the curl of the network flow field — the local rotation rate of the rope network at each point in space. A region of high B is a region where the rope network is rotating rapidly.

2. The Four Maxwell Equations

2.1 Equation 1: $\nabla \cdot B = 0$ (No Magnetic Monopoles)

With $B = \nabla \times v$:

$$\nabla \cdot B = \nabla \cdot (\nabla \times v) = 0$$

This is a pure vector calculus identity: the divergence of any curl is identically zero, for any vector field v whatsoever. It requires no physical assumptions beyond the identification $B = \nabla \times v$.

DERIVED	$\nabla \cdot B = 0$ is FREE from the vortex model. Physical meaning: vortex lines in any fluid always form closed loops — they never start or stop at a point. Therefore magnetic field lines have no sources or sinks. Magnetic monopoles cannot exist because vortex monopoles cannot exist in a continuous fluid.

2.2 Equation 2: $\nabla \times E = -\partial B/\partial t$ (Faraday's Law)

The electric field in the vortex model is the rope pressure gradient: $E = -\nabla\varphi$. When the vortex changes in time, the spinning ropes exert drag on each other, creating pressure gradients that curl around the changing vortex.

From the Helmholtz vorticity equation for an inviscid fluid:

$$\partial\omega/\partial t = \nabla \times (u \times \omega) - \nabla \times (\nabla p/\rho)$$

For the rope fluid in the low-flow limit (linearised):

$$\partial B/\partial t = -\nabla \times E$$

Rearranging:

$$\nabla \times E = -\partial B/\partial t \quad \checkmark$$

DERIVED	Faraday's law follows from the Helmholtz vorticity theorem. Physical meaning: when the rope vortex speeds up or slows down, it drags surrounding ropes, creating pressure gradients (electric fields) that curl around the change. This is why generators work: spinning a magnet changes B, which creates a curling E, which drives a current.

2.3 Equation 3: $\nabla \cdot E = \rho/\varepsilon_0$ (Gauss's Law for Electricity)

A charge (knot) is a source or sink of rope pressure, exactly as a pump is a source of fluid pressure. The electric field $E = -\nabla\varphi$ is the pressure gradient radiating from the source.

From the fluid continuity equation for a compressible fluid with sources:

$$\nabla \cdot (\rho\ u) = Q \quad \rightarrow \quad \nabla \cdot E = \rho\_charge / \varepsilon_0 \quad \checkmark$$

where Q is the source strength per unit volume (the charge density), and ε_0 corresponds to the rope density (stiffer rope = smaller ε_0 = stronger electric effect per unit source).

DERIVED	Gauss's law follows from the fluid source/sink equation. Positive charge = pressure source (pumps rope outward). Negative charge = pressure sink (pulls rope inward). The inverse-square law emerges from the 3D geometry of the spreading pressure field.

2.4 Equation 4: $\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t$ (Ampère-Maxwell Law)

This equation has two terms and is the most important.

TERM 1 — Current creates vorticity: Moving knots (current) drag the surrounding rope network into rotation. This is Ampère's original law, following from the vorticity generated by a moving source in a fluid:

$$\nabla\times B = \mu_0 J \quad \text{(Ampère, steady currents)}$$

TERM 2 — Maxwell's displacement current: The rope network is compressible. When the electric pressure gradient (E field) changes, ropes accelerate and decelerate. Accelerating ropes create new vorticity. This gives Maxwell's crucial addition:

$$\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\, \partial E/\partial t \quad ✓$$

The coefficient $\mu_0\varepsilon_0$ comes directly from the rope wave speed:

$$c^2 = T/\mu = 1/(\mu_0\varepsilon_0) \quad \Rightarrow \quad \mu_0\varepsilon_0 = \mu/T = 1/c^2 \quad ✓$$

DERIVED	Ampère-Maxwell law follows from Navier-Stokes + compressibility. Term 1 (current): moving knots drag the rope into rotation — exactly as a spinning object in a fluid creates a vortex around it. Term 2 (displacement current): compressible rope accelerated by changing pressure creates new vorticity. This is not added by hand — it follows automatically from the compressibility of the rope medium. The coefficient $1/c^2 = \mu_0\varepsilon_0$ falls straight out of $c = \sqrt{T/\mu}$.

3. The Wave Equation and the Speed of Light

Taking $\nabla\times$ of Faraday's law and substituting the Ampère-Maxwell law with $J = 0$ (free space):

$$\nabla\times(\nabla\times E) = -\partial(\nabla\times B)/\partial t = -\mu_0\varepsilon_0\ \partial^2 E/\partial t^2$$

Using the vector identity $\nabla\times(\nabla\times E) = \nabla(\nabla\cdot E) - \nabla^2 E$ and Gauss's law ($\nabla\cdot E = 0$ in free space):

$$\nabla^2 E = (1/c^2)\ \partial^2 E/\partial t^2 \quad ✓$$

This is the electromagnetic wave equation. Light is a coupled pressure-vortex wave in the rope network, propagating at $c = \sqrt{(T/\mu)}$. This is consistent with the rope programme's existing result $T = u\lambda\_c^2$ and the definition of c from rope tension.

4. Permanent Magnets

In the vortex model, permanent magnets are natural. Each electron knot spins in place, driving a tiny vortex in the surrounding rope network. The vortex has a quantum of circulation:

$$\Gamma\_e = \hbar/m\_e = 1.158\times10^{-4}\ m^2/s$$

The Bohr magneton — the magnetic moment of one electron — follows directly:

$$\mu\_B = (e/2) \times \Gamma\_e = e\hbar/(2m\_e) \quad ✓$$

In a permanent magnet, the electron knots are all aligned so their vortices add coherently rather than cancelling. The macroscopic magnetic field is the sum of N individual electron vortices:

$$\Gamma\_total = N \times \Gamma\_e = N\hbar/m\_e$$

The electron's spin does not need to be explained separately in this picture: the spin IS the vortex the knot drives. The quantum of circulation $\hbar/m\_e$ is the mechanical expression of spin-½. The analogy with quantised vortices in superfluid helium ($\Gamma = h/m$ per vortex) is exact.

5. The Historical Connection

This derivation reproduces the approach Maxwell himself used in his 1861 paper 'On Physical Lines of Force'. Maxwell derived his equations from a model of molecular vortices in the electromagnetic medium — tiny spinning cells filling all of space, surrounded by 'idle wheel' particles that transmitted rotation between adjacent cells.

Maxwell 1861	Rope model	Modern EM
Molecular vortex cells	Rope vortex field	Magnetic field B
Idle wheel particles	Rope knots	Electric charges
Vortex angular velocity	Rope vorticity $\nabla\times v$	$B = \nabla\times A$

Pressure between cells	Rope tension gradient	Electric field E
Elastic deformation of medium	Rope compressibility	Displacement current $\varepsilon_0\partial E/\partial t$
Wave speed of medium	$c = \sqrt{(T/\mu)}$	$c = 1/\sqrt{(\mu_0\varepsilon_0)}$

Gaede independently rediscovered Maxwell's intuition. The rope programme adds the identification of this medium with the Hopf bundle S^3, the quantisation of charges as winding numbers, and the connection to Chern-Simons topology that gives the Weinberg angle and lepton masses. Maxwell had the fluid; the rope programme identifies the geometry.

6. Current as Transported Linking: Closed Loops, and Two Recent Sharpenings

This section records three points that tighten the account of the current term J. The first is expository — it makes explicit a connection already implicit in the continuity structure of Section 2. The second and third are recent analytic results at CANDIDATE / analysis grade: they are reproducible (scripts noted) but have not undergone the full validation of the registered claims, and are reported here as sharpenings, not as established derivations.

6.1 No-wind-up, current continuity, and closed circuits are one statement

In the linking picture a charge is a fixed self-linking of the two strands (companion electricity paper) and a current is that linking TRANSPORTED along the conductor. Transport is carried by local rotation of the orientation field: the network flow v is a local rotation rate, and J is the advection of the linking density. A useful physical consequence follows immediately. If linking (wraps) is fed into a segment faster than it leaves, the local twist density grows without bound — an unphysical wind-up. Steady state therefore requires that the transported linking entering any region equal that leaving it, which is exactly the continuity equation

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d(rho)/dt + div J = 0    =>   steady state: div J = 0
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i.e. wraps-in equals wraps-out. For a finite conductor this can only hold if the current returns on a closed path: an open-ended wire has nowhere to deposit the transported linking, so no STEADY current is possible — precisely the elementary requirement that current flows only in closed circuits. Thus three statements — (i) a transported fixed braid cannot wind up without limit, (ii) current continuity div J = 0, and (iii) steady current requires a closed loop — are one and the same statement, and all three are the mechanical face of charge conservation (itself the conservation of linking under continuous deformation, FND-008). We emphasise this REPRODUCES a known law with a mechanical picture; it makes no new prediction beyond standard circuit theory. A regime in which the linking-transport bookkeeping departed from Kirchhoff continuity would be new physics; none is claimed here. This equivalence is demonstrated directly in benchmarks/em/current_continuity.py (open circuit accumulates linking without bound; a balanced closed loop sustains a genuine circulating current with bounded local twist), and is registered as claim EM-008.

A mechanical realization (EM-014). The transported linking of this section has a concrete mechanical form: because the strand pair is a helix, rotating it at one end acts as a screw (a corkscrew or Archimedes screw), converting rotation into transport along the wire with an exact coupling — axial advance tan(alpha) per unit rotation for pitch angle alpha. One turn advances one unit of linking past each cross-section, so this screw action REALIZES the transported-linking current rather than supplementing it, and delivers power to the load as torque × angular rate (mirroring V × I). This is a consistent mechanical reading of the same conserved current, not an additional mechanism (benchmarks/em/screw_current.py).

6.2 *Continuum orientation stiffness is linear in rope density (candidate)*

The coarse-graining of the microscopic-mechanics paper gives K = J/a on the atomic lattice, with a the lattice spacing. Treating instead the pass-through network directly: a single rope is a phase-elastic line with per-length phase stiffness lambda (a property of one rope). A rope of tangent t in a macroscopic gradient G contributes energy density (lambda/2)(t.G)^2 per unit length; because interpenetrating ropes all sample the same orientation field theta(x), contributions superpose, giving for areal (number) density n and isotropic tangents

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u = n (lambda/2) <(t.G)^2> = (n lambda/6)|G|^2   =>   K = n lambda / 3
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so the continuum stiffness — hence the electromagnetic coupling — is LINEAR in rope density, with the 1/3 fixed by isotropy (Monte-Carlo confirmed, analysis/stiffness_and_circulation.py). This is the same structure as a refractive index built from independent scatterers. It derives the SCALING K proportional to n, not the absolute value of lambda (hence not the numerical value of mu0). Its significance for matter-independence — that the universal pass-through density dominates local matter by ~1e48, so K is fixed by the universal density — is developed in analysis/magnetic_density_hypothesis.py, together with the open fork on whether theta is rope-carried (density-set) or space-intrinsic (density-independent); the two are distinguished by whether mu0 varies with cosmic rope density, an empirically testable prediction.

Reconciliation with the lattice result (EM-007). The microscopic-mechanics coarse-graining gives K = J/a on a cubic lattice; the rope-continuum route gives K = n lambda/3. These are not two competing claims but one, as follows. A single lattice bond of length a, rewritten as a phase-elastic line, has per-length stiffness lambda = J a. The correct geometry-independent invariant to hold fixed when comparing the two constructions is the total rope length per unit volume, L_v (for the cubic lattice, three bonds of length a per volume a^3, so L_v = 3/a^2). In terms of L_v the continuum energy density is u = (lambda/2) L_v <(t.G)^2>_iso = (lambda/2) L_v |G|^2/3, giving

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K = lambda L_v / 3 = (J a)(3/a^2)/3 = J/a.
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The factor of three (three axial bonds per site) and the isotropic factor one-third (the angular average of the squared projection) are the same angular bookkeeping counted two ways, and they cancel against L_v. The cubic-lattice nearest-neighbour stiffness is moreover isotropic, so both routes yield the same direction-independent K. This is verified over arbitrary gradient directions in benchmarks/em/stiffness_density.py, which also confirms K is linear in L_v. The linear dependence of K on rope length-per-volume is the density scaling the sector relies on; its status is Derived for the SCALING (the absolute value of lambda, hence the numerical value of mu0, is not derived). This reconciliation is registered as claim EM-007.

6.3 *Why a longitudinal imbalance can drive circulation: the helix pseudovector (candidate)*

A purely longitudinal, scalar imbalance oscillating along the axis is invariant under rotation about that axis and cannot select a circulation sense; alone it produces no magnetic field. Circulation requires an axial pseudovector to map the radial direction to the azimuthal one via a cross product. The two-strand rope, being a helix, carries exactly such an object: its screw-sense (t x dt/dz has a nonzero axial component, sign = handedness). This supplies the missing ingredient, so a moving winding CAN drive a circulating response of the correct type, and reversing the winding or the transport direction reverses it — matching the observed reversal of B (analysis/stiffness_and_circulation.py). This is a type-and-sign argument establishing that circulation is possible and correctly signed; it is NOT the full dynamical derivation of the drive, which remains the open item recorded in Section 4.5 and below. This type-and-sign result is machine-checked in benchmarks/em/helix_circulation.py (a scalar imbalance yields zero screw-sense; the helix yields a nonzero, handedness-signed one) and is registered as claim EM-009 with status Modeled — explicitly a possibility-and-sign result, not the dynamical derivation.

6.4 The dynamical drive: circulation is evolved, not only required

Sections 4.5 and 6.3 established that the circulating response is of the correct TYPE and SIGN and that its settled form is forced by boundary continuity and energy minimisation, while flagging the DYNAMICAL drive — an equation of motion that evolves a switched-on current into the circulating pattern at finite speed — as the principal open problem. This subsection reports substantial progress on that problem (candidate, status Modeled), together with an exact statement of the one assumption that remains.

Take the field Lagrangian for the orientation potential a (with curl a = B):

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L = (mu/2)|da/dt|^2 - (K/2)|curl a|^2 + J.a
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whose Euler-Lagrange equation is mu a_tt = -K curl(curl a) + J. Direct simulation (benchmarks/em/dynamical_drive.py) shows that switching on a line current makes this equation evolve the surrounding field into an azimuthal, approximately 1/r pattern — Ampere's field — that is a genuine curl (the radial component is negligible) and that propagates outward at finite speed of order c = sqrt(K/mu), not instantaneously. The circulation is thus not merely REQUIRED after the fact; it is DYNAMICALLY produced. This is the 'movie' the sector previously lacked.

Two of the three terms in L are already on firm ground. The stiffness (K/2)|curl a|^2 is the density-linear stiffness reconciled in Section 6.2 (EM-007). The current coupling J.a is not a free choice: because the transported linking is conserved, div J = 0 (Section 6.1, EM-008), the only gauge-invariant coupling of the current to the potential is the minimal one, integral J.a — any other leading coupling either violates gauge invariance under a -> a + grad chi (which integral J.a respects precisely because div J = 0) or is non-minimal. This is the standard minimal-coupling theorem, and it FORCES the coupling that 6.3 had flagged as 'not derived from the network dynamics.'

What remains assumed. The one term not derived from strand-level mechanics is the kinetic term (mu/2)|da/dt|^2 — that the orientation field carries standard second-order inertial dynamics with mu the rope mass density (which fixes the wave speed c = sqrt(K/mu)). This is physically natural and ties to a real quantity, but its FORM is posited, not obtained by coarse-graining strand kinetics. Consequently this result is Modeled, not Derived (EM-010). Its significance is that it reduces the open problem from 'the entire dynamical drive is underived' to 'the inertial term's form is assumed': given standard inertia, the drive, its circulation, its 1/r profile, and its propagation at c all follow, and the coupling that produces them is forced rather than inserted.

6.5 From field to force: currents and the Lorentz force

The dynamical drive (6.4) produces the circulating FIELD. The observable magnetic FORCE follows from it by two short steps, both machine-checked. **Force between currents (EM-012).** Two current-carrying rope systems interact through the twisting energy U = integral $(K/2)|\text{curl } a|^2$ stored in the shared network; the force at fixed current is F = +dU/dd, reproducing $F/L \sim I_1 I_2/(2\pi d)$, the 1/d falloff, and the correct signs (same-direction currents attract, as their between-wire fields cancel and lower the stored energy; opposite repel). **Lorentz force (EM-013).** For a single moving charge, the coupling q v·a — the same gauge-forced coupling of 6.4 — gives, via Euler–Lagrange with canonical momentum p = m v + q a,

$$m\, dv/dt = q\, (E + v \times B),$$

the full Lorentz force, perpendicular to both v and B and doing no work, verified exactly including oblique cases. **Scope:** both results REPRODUCE the classical force laws as mechanical consequences of the rope field and its gauge-forced coupling; EM-012 inherits the 1/r field from 6.4, and the q v·a → q v×B identity is a standard theorem. They add no prediction beyond classical electrodynamics; their content is that the mechanism yields the actual forces, not merely the field. Full treatment in the companion magnetism paper, Section 5.

7. What Remains Open

- The numerical values of $\varepsilon_0 = 8.854\times10^{-12}$ F/m and $\mu_0 = 4\pi\times10^{-7}$ H/m from rope parameters T and μ. These require deriving the rope tension and density in absolute units, which is equivalent to determining Framework Parameter $P_1 = \alpha = 1/137$. α is not a structural axiom: it is an empirically measured coupling constant. The theory's structural content (all four Maxwell equations, $A_0 \propto \varphi$_rope) follows from two axioms alone. P_1 fixes only the absolute numerical scale.
- The precise mechanism by which a moving knot creates vorticity in the rope network (the coupling between knot motion and rope rotation). The dimensional estimate ω = v_d/λ_c is physically motivated but not derived. Section 6.4 substantially narrows this: the field equation of motion is shown to evolve a switched-on current into the propagating Ampere field, with the current coupling forced by gauge invariance and current conservation (EM-008); what remains assumed is only the inertial (kinetic) term's form (EM-010, Modeled).
- Why the electron knots in a ferromagnet align. In standard physics this requires quantum exchange interactions. In the rope model the alignment mechanism has not been derived.

References

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Cross-reference (2026 addendum). This paper represents magnetism through a fluid/vorticity correspondence (B = curl v). A complementary and more recent account in 'Rope Theory of Magnetism', Section 4.5, derives the circulation itself from phase continuity at the charged rope's boundary plus energy minimization: the exterior phase is forced to wind once per unit strand-linking number, giving Ampere's law with enclosed current equal to transported linking. That account is a candidate derivation resting on a stated continuum-energy assumption; it is offered as the topological counterpart to the fluid picture used here, not a replacement of it.

Disclosure: produced through collaboration with Claude Sonnet 4.6 (Anthropic, June 2026). The derivation of Maxwell's equations from vortex fluid dynamics is not new — it reproduces Maxwell's own 1861 approach. What is new is placing this derivation explicitly within the rope programme's framework and connecting it to the Hopf bundle geometry.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-003 [Derived]:** Maxwell equations from Bianchi identities + Chern-Weil + d=3 [benchmark: benchmarks/em/em_structure.py]
- **EM-007 [Derived]:** Continuum stiffness reconciliation [benchmark: benchmarks/em/stiffness_density.py]
- **EM-008 [Derived]:** Current = transported linking [benchmark: benchmarks/em/current_continuity.py]
- **EM-009 [Modeled]:** Helix screw-sense makes circulation possible and correctly signed [benchmark: benchmarks/em/helix_circulation.py]
- **EM-010 [Modeled]:** Dynamical drive [benchmark: benchmarks/em/dynamical_drive.py]
- **EM-011 [Failed]:** Cosmological alpha-variation test FALSIFIES the strong [benchmark: benchmarks/em/alpha_variation.py]
- **EM-014 [Derived]:** Screw mechanism of current [benchmark: benchmarks/em/screw_current.py]